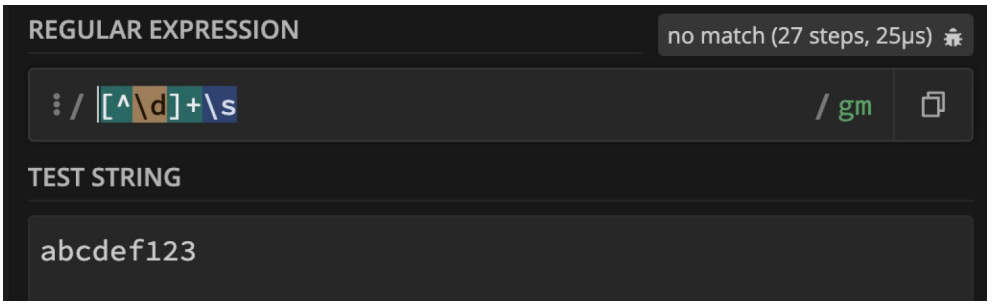


Figure 8-3. *Interactive regex editor before…*

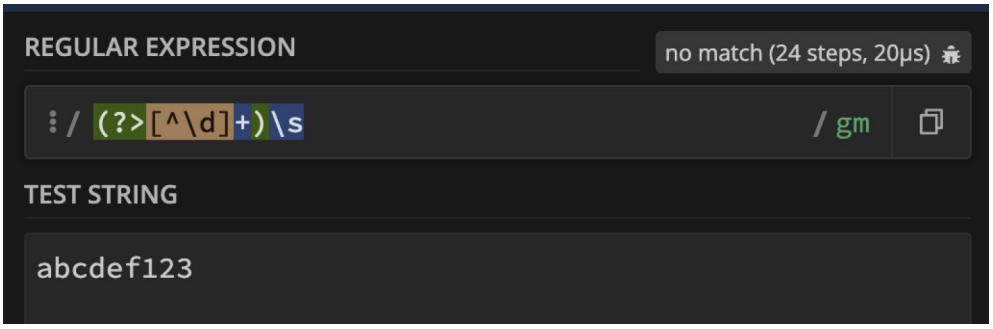


Figure 8-4. *Interactive regex editor after…*

Lookahead, Lookbehind

Let's return to the idea of consuming characters when matching. The normal operation of the PCRE engine is to move along the sample string as it matches according to the expression. If we have a simple pattern like `/abc(\w+)/` and a sample of `'abcdef'`, then the engine moves along the string one character at a time, matching first `a`, `b`, and `c`, then the remaining word characters of `def` (and in this case creating the `\1` capture group, which contains `def`). In some cases, it is desirable to match without consuming, and for this we have *lookahead* and *lookbehind* assertions (collectively known as *lookaround*). A lookaround will check that the sample matches a sub-pattern without changing positions in the sample; they're another form of zero-width assertion.

The idea of looking ahead (or behind) but not consuming any part of the sample is subtle one, which can be explained with the following example. Imagine that you wanted to assert that one letter was not followed by another in a particular word; this would perhaps seem straightforward at first glance. We know how to use character type escapes